

AI & Data Science Knowledge Base

A Sample Document for the RAG Document Assistant Project

This fictional knowledge base is created specifically for testing a Retrieval-Augmented Generation (RAG) application. It contains structured information that can be retrieved and used to answer questions.

1. Company Overview

NovaMind Analytics is a fictional technology company focused on artificial intelligence, data science, business intelligence, and automation. The company helps organizations turn business data into useful insights and build practical AI-powered applications.

The company has four primary teams: Data Analytics, Machine Learning, AI Applications, and Business Intelligence. The Data Analytics team focuses on data cleaning, exploratory analysis, and reporting. The Machine Learning team develops predictive models. The AI Applications team builds natural language and generative AI solutions. The Business Intelligence team creates dashboards and KPI reporting systems.

2. Core Technologies

The main programming language used by the engineering teams is Python. SQL is used for querying and managing relational data. Power BI is used for interactive business dashboards. Common Python libraries include pandas, NumPy, scikit-learn, PyTorch, TensorFlow, and Streamlit.

For natural language processing applications, the teams may use tokenization, text preprocessing, embeddings, vector databases, retrieval systems, and large language models. For computer vision applications, common tasks include image classification, object detection, and image preprocessing.

3. Artificial Intelligence Projects

One example project is a Financial Phishing Email Detector. The system analyzes email text and predicts whether an email is likely to be legitimate or phishing. A typical pipeline includes text cleaning, TF-IDF feature extraction, model training, evaluation, and a web interface.

Another example is a Waste Image Classification system. The application classifies waste images into categories such as cardboard, glass, metal, paper, plastic, and general waste. A convolutional neural network can be used to learn visual patterns from training images.

4. RAG Systems

Retrieval-Augmented Generation, commonly called RAG, combines information retrieval with text generation. Instead of relying only on information learned during model training, a RAG application retrieves relevant content from a connected knowledge source and provides that content to a language model as context.

A basic RAG workflow contains five important stages: document loading, text splitting, embedding generation, vector storage and retrieval, and answer generation. Documents are divided into smaller chunks so that relevant sections can be retrieved efficiently.

Embeddings represent text as numerical vectors. Similar pieces of text have relatively similar vector representations. A vector database such as FAISS can store these vectors and perform similarity searches.

5. Internship Program

NovaMind Analytics offers a fictional eight-week AI and Data Science internship program. The program introduces participants to Python, SQL, data analysis, machine learning, natural language processing, and deployment basics.

During the internship, participants complete weekly exercises and one final project. The final project should include data preparation, model or application development, evaluation, documentation, and a simple user interface.

6. Frequently Asked Questions

What is RAG? RAG stands for Retrieval-Augmented Generation. It retrieves relevant information and supplies it to a language model before generating an answer.

Why use embeddings? Embeddings convert text into numerical representations that allow semantic similarity searches.

What is FAISS? FAISS is a library designed for efficient similarity search over vectors.

What language is mainly used? Python is the primary programming language used by the engineering teams.

What is the phishing detector designed to identify? It predicts whether an email is likely to be legitimate or phishing.

What does the waste classifier do? It classifies waste images into predefined categories.